

story about your coworker Bob, but for that story to make sense, you first have to explain something about your cousin Carol. You finish your story about Carol and go back to talking about Bob, and when you finish your story about Bob, you go back to talking about Alice. Then you are reminded about your brother David, so you tell a story about him. Eventually, you get around to finishing your original story about Alice.

Your conversation follows a stack-like structure, as in Figure 1-5. The conversation is stack-like because the current topic is always at the top of the stack.

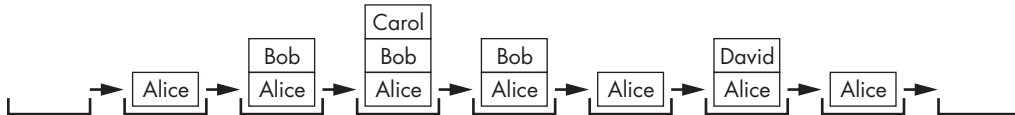


Figure 1-5: Your meandering conversation stack

In our conversation stack, the new topics are added to the top of the stack and taken off as they are completed. The previous topics are “remembered” underneath the current topic in the stack.

We can use Python lists as stacks if, to amend the list’s contents, we limit ourselves to the `append()` and `pop()` methods to perform pushing and popping. JavaScript arrays can also be used as stacks through their `push()` and `pop()` methods.

**NOTE** *Python uses the terms `list` and `item`, while JavaScript uses the terms `array` and `element`, but they are respectively identical for our purposes. In this book, I use the terms `list` and `item` for both languages.*

For example, consider this `cardStack.py` program, which pushes and pops string values of playing cards to the end of a list named `cardStack`:

---

```

Python cardStack = ❶ []
❷ cardStack.append('5 of diamonds')
  print(', '.join(cardStack))
  cardStack.append('3 of clubs')
  print(', '.join(cardStack))
  cardStack.append('ace of hearts')
  print(', '.join(cardStack))
❸ cardStack.pop()
  print(', '.join(cardStack))
  
```

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The following `cardStack.html` program contains the equivalent code in JavaScript:

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```

JavaScript <script type="text/javascript">
  let cardStack = ❶ [];
  ❷ cardStack.push("5 of diamonds");
  document.write(cardStack + "<br />");
  cardStack.push("3 of clubs");
  document.write(cardStack + "<br />");
  
```

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